

CHAPTER 2: LITERATURE REVIEW

2.1 Overview of Big Data Security Challenges

The rapid expansion of enterprise data environments has fundamental transformed modern organizational decision-making processes, yet it has simultaneously exposed critical vulnerabilities in large scale data processing infrastructures. Big Data architectures are inherently characterized by the core dimensions of Volume, Velocity, Variety, Veracity and Value. While these characteristics enable unprecedented real-time analytical capabilities, they introduce complex security paradigms that traditional perimeter-based security measures fail to address adequately. In high volume streaming environments, traditional access control mechanisms and single node inspection firewalls become severe processing bottlenecks, leading organizations to balance performance against rigorous security controls.

Threat actors actively exploit the decentralized nature of distributed nodes through sophisticated attack vectors, including unauthorized data access, privilege escalation and side channel data leakage. Furthermore, data ingestion pipelines frequently combine structured, semi structured and unstructured data from heterogeneous sources, which complicates the uniform enforcement of encryption and data masking policies. The fundamental conflict in Big Data security revolves around maintaining data utility for analytical algorithms while upholding strict data confidentiality and privacy guarantees. As enterprise architectures transition toward multi tenant cloud ecosystems, risks surrounding data provenance, unauthorized reidentification and regulatory compliance breaches expand exponentially, necessitating structural security integration directly into distributed storage engines and processing frameworks.

2.2 Security Architectures in Distributed File Systems

Distributed File Systems, such as the Hadoop Distributed File System, Apache Ozone and cloud native Object Stores like Amazon S3, form the foundational storage infrastructure for modern Big Data analytical clusters. However, their core architecture, which relies on replicating data blocks across geographically or logically distributed physical DataNodes, introduces distinct security challenges. Security architectures in distributed storage environments must ensure robust user authentication, fine grained access authorization, transparent data at rest encryption and wire level transport security without incurring acceptable latencies or computational bottlenecks.

Modern distributed file systems integrate multi-layered security frameworks to mitigate these risks. Kerberos is widely deployed to provide strong mutual authentication between client applications, NameNodes and DataNodes, preventing impersonation attacks. Authorization is enforced using standard File System Access Control Lists combined with centralized policy engines, such as Apache Ranger or Apache Sentry, to maintain consistent rule enforcement across complex multi tenant clusters. To protect sensitive data at rest, Transparent Data Encryption isolates sensitive storage zones and encrypts individual data blocks using key management servers. However, applying cryptographic operations at scale introduces significant performance overhead, where symmetric cryptographic primitives provide hardware accelerated throughput, whereas asymmetric and advanced homomorphic primitives introduce severe latency constraints.

Encryption Framework / Mechanism	Security Level & Guarantee	Throughput Impact	Latency Overhead	Computational Complexity	Primary Use Case in DFS
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AES 256 GCM (Symmetric TDE)	High (FIPS 140-2 compliant)	Minimal degradation (~3–7%)	Low (< 2 ms)	$O(N)$ (Hardware accelerated via AES NI)	Data at Rest Encryption for HDFS/S3 blocks
Attribute Based Encryption (ABE / CP-ABE)	High (Fine grained access control)	Moderate reduction (~18–25%)	Medium (~15–35 ms)	$O(N \cdot A)$ where $ A $ is attribute count	Fine grained multi tenant access in shared DFS
Partial Homomorphic Encryption (PHE - Paillier)	High (Secures mathematical operations)	High degradation (~50–65%)	High (~120–250 ms)	$O(N^3)$ (Large integer exponentiation)	Encrypted aggregations over numeric data blocks
Fully Homomorphic Encryption (FHE - CKKS/BFV)	Extreme (Secures arbitrary computation)	Severe degradation (> 90%)	Very High (> 1500 ms)	$O(N \log N)$ over polynomial rings	Secure privacy preserving cloud computations

Table 2.1: Comparison of Encryption Overhead Across Distributed Systems

2.3 Summary Matrix of Past Research

Academic literature surrounding Big Data privacy models has evolved through distinct technological phases, transitioning from traditional heuristic anonymization algorithms to rigorous mathematical definitions and decentralized machine learning frameworks. Early methodologies prioritized identity protection in static relational databases, whereas modern solutions focus on dynamic data streams and distributed machine learning pipelines. The comparative matrix below synthesizes benchmark literature, detailing the underlying methodologies, evaluation metrics, core contributions, and technical limitations of existing privacy models.

Author(s) & Year	Privacy Model / Technical Domain	Methodology & Mechanism Proposed	Key Evaluation Metrics	Key Findings & Contributions	Identified Research Limitations
Sweeney (2002)	k-Anonymity / Data Anonymization	Quasi identifier suppression and value generalization.	Suppression Rate, Utility Loss Ratio	Established foundation for Privacy Preserving Data Publishing (PPDP).	Vulnerable to background knowledge and homogeneity attacks.
Machanavajjhala et al. (2007)	l-Diversity / Attribute Privacy	Enforces diversity of sensitive attributes within equivalence classes.	Entropy, Feature Correlation Preservation	Resolved identity disclosure issues present in basic k-anonymity.	Suffers from high utility loss vulnerable to skewness/similarity attacks.

Li et al. (2007)	t-Closeness / Distribution Alignment	Equivalence class distribution matching using Earth Mover's Distance.	Earth Mover's Distance (EMD), Information Loss	Prevented attribute disclosure in dense categorical datasets.	High computational complexity scales poorly to streaming Big Data.
Dwork (2008)	Differential Privacy (epsilon-DP)	Laplace / Gaussian noise injection into query outputs.	Privacy Budget (epsilon), Query Error Rate, Accuracy	Provided a mathematically provable privacy guarantee independent of attacker knowledge.	Sharp trade off between privacy budget epsilon and overall data utility.
Erlingsson et al. (2014)	Local Differential Privacy (LDP)	Client side Randomized Response (RAPPOR framework).	False Positive Rate, Heavy Hitter Detection Accuracy	Allowed privacy preserving telemetry collection without a trusted server.	High noise volume requires massive sample sizes for statistical accuracy.
McMahan et al. (2017)	Privacy Preserving Machine Learning	Federated Learning combined with Differential Privacy (DP-FL).	Model Convergence Time, Top 1 Accuracy, ϵ -Budget	Enabled collaborative model training without centralizing raw user data.	High communication overhead vulnerable to model inversion attacks if epsilon is too high.
Zhang et al. (2022)	Dynamic Big Data Streaming Privacy	Adaptive LDP budget allocation for high velocity streams.	Latency (ms), Precision/Recall, Privacy Loss Curve	Reduced latency in streaming data processing while keeping privacy limits intact.	Parameter tuning requires prior knowledge of data distribution limits

2.4 Research Gap

A comprehensive synthesis of existing literature reveals several critical research gaps in the domain of Big Data privacy protection. The primary research gap lies in the persistent utility privacy imbalance encountered when processing high dimensional data streams. Conventional Local Differential Privacy frameworks apply uniform noise injection mechanisms across all data attributes regardless of their underlying analytical significance. When scaled to continuous, high velocity streaming datasets, this indiscriminate noise addition severely degrades statistical utility and destroys non linear feature correlations that are essential for accurate real-time machine learning and analytics.

Furthermore, current privacy implementations suffer from rigid privacy budget allocation models. In real world multi query environments, systems rely on static privacy budgets that accumulate consumption exponentially over continuous queries. Once the privacy budget is fully depleted,

systems are forced to either halt analytical query execution entirely, causing service disruptions, or proceed with unbudgeted queries that expose individual records to privacy reconstruction attacks. Existing literature lacks dynamic mechanisms capable of adjusting privacy parameters based on real time threat assessments, query contexts and changing data distributions.

Lastly, severe architectural scalability bottlenecks remain unaddressed in distributed processing environments. Advanced cryptographic paradigms such as Fully Homomorphic Encryption demand immense computational overhead, resulting in processing latencies that are impractical for production Big Data workloads. On the other hand, classic heuristic anonymization approaches scale with quadratic computational complexity, making them inefficient for massive, unstructured storage engines like distributed object stores. There remains a distinct lack of scalable, lightweight privacy frameworks designed specifically to balance formal mathematical privacy guarantees with high throughput execution in distributed file systems.